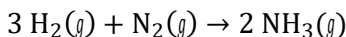


Unit 5 Multiple Choice Practice

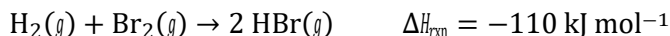


1. Gaseous hydrogen and nitrogen combine in the reaction above to form ammonia, NH_3 . Using the table of entropies provided below, calculate the standard entropy change, $\Delta S^\circ_{\text{rxn}}$, for the reaction shown at 25°C .

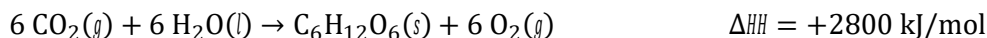
Substance	S° at 25°C (J/mol K)
$\text{H}_2(g)$	131
$\text{N}_2(g)$	192
$\text{NH}_3(g)$	193

- a. -199 J/mol K b. -386 J/mol K c. $+199 \text{ J/mol K}$ d. $+386 \text{ J/mol K}$
2. Which of the following is true for an endothermic reaction?
- The activation energy is always greater than the activation energy for an exothermic reaction.
 - Energy is released over the course of the reaction.
 - The strength of the bonds in the reactants exceeds the strength of the bonds in the products.
 - The products have more thermal energy than the reactants.

3. If the standard enthalpy of formation for $\text{HBr}(g)$ is -35 kJ mol^{-1} , what is ΔH°_f for $\text{Br}_2(g)$?



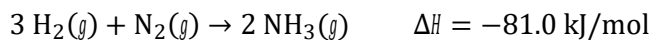
- a. 0 kJ mol^{-1} b. $-40. \text{ kJ mol}^{-1}$ c. $+40. \text{ kJ mol}^{-1}$ d. $+75 \text{ kJ mol}^{-1}$
4. Consider the following reaction showing photosynthesis:



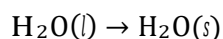
How much thermal energy is absorbed or released when 2.0 moles of glucose, $\text{C}_6\text{H}_{12}\text{O}_6(s)$, is metabolized in the presence of excess oxygen gas, releasing CO_2 and H_2O ?

- a. 2800 kJ absorbed b. 5600 kJ absorbed c. 2800 released d. 5600 kJ released

Use the following information to answer the next three questions.

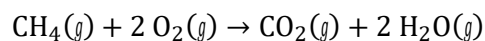


5. Gaseous hydrogen and nitrogen combine in the reaction above to form ammonia, NH_3 . What is the value of the heat of formation of $\text{NH}_3(g)$?
- a. -40.5 kJ/mol b. -81.0 kJ/mol c. $+40.5 \text{ kJ/mol}$ d. $+81.0 \text{ kJ/mol}$
6. If 28.0 g of $\text{N}_2(g)$ is combined with 28.0 g of $\text{H}_2(g)$ in a sealed flask and the reaction is allowed to go to completion, what chemical species will still be present inside of the flask?
- a. $\text{NH}_3(g)$ only c. $\text{NH}_3(g)$ and $\text{H}_2(g)$
 b. $\text{NH}_3(g)$ and $\text{N}_2(g)$ d. $\text{NH}_3(g)$, $\text{N}_2(g)$ and $\text{H}_2(g)$
7. If 28.0 g of $\text{N}_2(g)$ is combined with 28.0 g of $\text{H}_2(g)$ in a sealed flask and the reaction is allowed to go to completion, how much heat is absorbed or released?
- a. 81.0 kJ absorbed b. 375 kJ absorbed c. 81.0 kJ released d. 375 kJ released



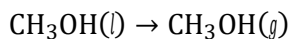
8. Is the process above endothermic or exothermic? Why?
- a. Exothermic, because forming new intermolecular attractions between water molecules releases energy.
 b. Endothermic, because energy is needed to form new intermolecular attractions between water molecules.
 c. Exothermic, because forming new bonds between water molecules releases energy.
 d. Endothermic, because energy is needed to form new bonds between water molecules.

Bond	Average Bond Dissociation Energy (kJ/mol)
C – H	415
O = O	495
C = O	798
O – H	463



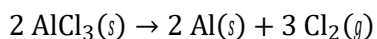
9. Use the information provided above to calculate the molar enthalpy of combustion, $\Delta H^\circ_{\text{comb}}$, for methane gas.

- a. -798 kJ mol^{-1} b. $-1226 \text{ kJ mol}^{-1}$ c. $+798 \text{ kJ mol}^{-1}$ d. $+1226 \text{ kJ mol}^{-1}$



10. For the boiling of methanol, CH_3OH , $\Delta H^\circ = +37.6 \text{ kJ mol}^{-1}$ and $\Delta S^\circ = +111 \text{ J mol}^{-1} \text{ K}^{-1}$. Calculate the standard free energy change of this reaction, $\Delta G^\circ_{\text{rxn}}$.

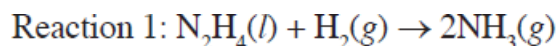
- a. -4.5 kJ/mol b. -2.8 kJ/mol c. $+2.8 \text{ kJ/mol}$ d. $+4.5 \text{ kJ/mol}$



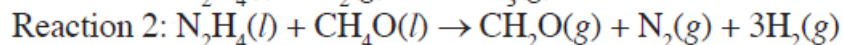
11. The reaction above is spontaneous under standard conditions, but it becomes thermodynamically unfavorable as temperature decreases. Which of the following is true?

- a. ΔS and ΔH are both negative. c. ΔS is negative, and ΔH is positive.
 b. ΔS and ΔH are both positive. d. ΔS is positive, and ΔH is negative.

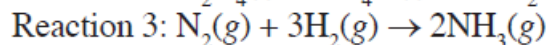
Use the following information to answer the next two questions.



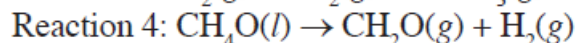
$$\Delta H = ?$$



$$\Delta H = -37 \text{ kJ/mol}_{\text{rxn}}$$

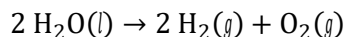


$$\Delta H = -46 \text{ kJ/mol}_{\text{rxn}}$$



$$\Delta H = -65 \text{ kJ/mol}_{\text{rxn}}$$

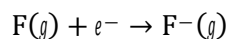
12. If reaction 4 were repeated at a higher temperature, how would the reaction's value for ΔG be affected?
- It would become more negative because entropy is a driving force behind this reaction.
 - It would become more positive because enthalpy is a driving force behind this reaction.
 - It will stay the same because reaction 4 is never thermodynamically favorable.
 - It will stay the same because reaction 4 is thermodynamically favorable at all temperatures.
13. Under what conditions would reaction 2 be thermodynamically favored?
- It is always favored.
 - It is never favored.
 - It is only favored at low temperatures.
 - It is only favored at high temperatures.



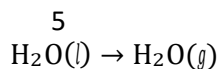
14. Based on the information in the table below, calculate $\Delta H^\circ_{\text{rxn}}$ for the reaction shown above.

Bond	Average Bond Dissociation Energy (kJ/mol)
H – H	430
O = O	500
O – H	460

- -900 kJ mol^{-1}
- -480 kJ mol^{-1}
- $+480 \text{ kJ mol}^{-1}$
- $+900 \text{ kJ mol}^{-1}$



15. Is the process above endothermic or exothermic? Why?
- Exothermic, because of the attraction between $\text{F}(g)$ and an electron.
 - Endothermic, because of the attraction between $\text{F}(g)$ and an electron.
 - Exothermic, because the process decreases the total number of particles from two to one.
 - Endothermic, because the process decreases the total number of particles from two to one.



16. Which of the following is true for the above reaction?

- a. The value of ΔS is negative.
- b. The value of ΔG is negative at 298 K.
- c. The value of ΔH is positive.
- d. The reaction is favored at 1.0 atm and 298 K.

Use the following information to answer the next two questions.

When ammonium nitrate (NH_4NO_3) dissolves in water, the temperature of the water decreases dramatically.

17. During this reaction, energy transfers from:

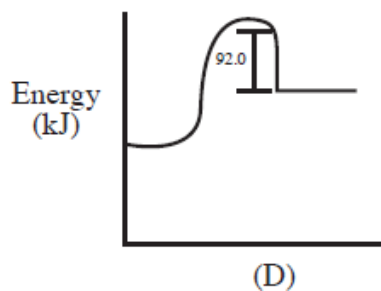
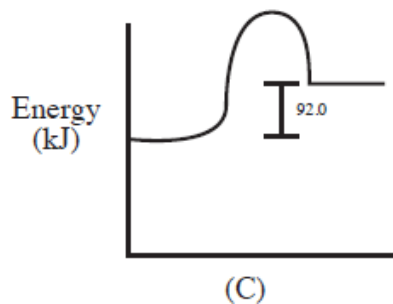
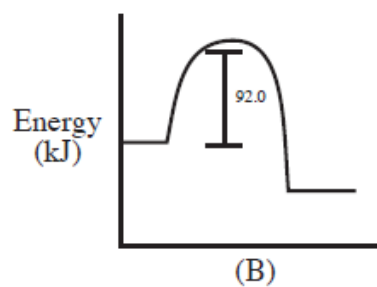
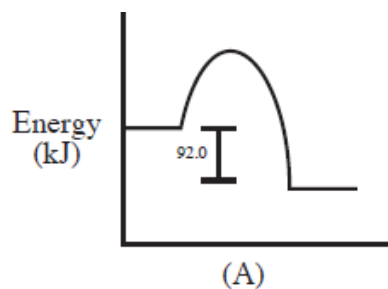
- a. the reactants to the products.
- b. the reactants to the system.
- c. the system to the surroundings.
- d. the products to the surroundings.

18. Which is the primary driving factor behind this reaction?

- a. Entropy
- b. Enthalpy
- c. Both enthalpy and entropy
- d. Neither enthalpy and entropy



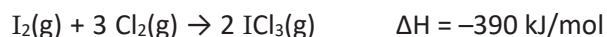
19. Which of the following energy level diagrams gives an accurate representation of the above reaction?





20. Regarding the reaction above, how much heat is absorbed or released when 1.0 mol of solid carbon reacts with 1.0 mol of solid sulfur to produce $\text{CS}_2(\text{l})$?

- a. 46 kJ of heat is released.
- b. 46 kJ of heat is absorbed.
- c. 92 kJ of heat is released.
- d. 92 kJ of heat is absorbed.



21. Which of the following statements accurately describes the above reaction?

- a. The entropy of the products exceeds that of the reactants.
- b. $\text{I}_2(\text{g})$ will always be the limiting reagent.
- c. This reaction is never thermodynamically favored.
- d. The temperature of the surroundings will increase as this reaction progresses.

22. Inside a calorimeter, 100. mL of 1.0 M formic acid (HCOOH), a weak acid, and 100. mL of 2.0 M sodium hydroxide, NaOH , are mixed. The specific heat of the mixture is approximately $4.2 \text{ J/g}^\circ\text{C}$, and the density is identical to that of water. If the temperature of the combined solution rises 8.0°C during the course of the reaction, calculate the approximate amount of heat released.

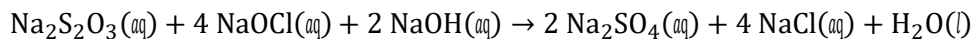
- a. 0.84 kJ
- b. 3.4 kJ
- c. 6.7 kJ
- d. 11.8 kJ

23. Consider the previous question. If the experiment is repeated with 150. mL of 1.0 M HCOOH and 50.0 mL of 2.0 M NaOH , what would happen to the values for ΔT and ΔH_{rxn} ?

	ΔT	ΔH_{rxn}
(A)	Increase	Increase
(B)	Stay the same	Stay the same
(C)	Decrease	Stay the same
(D)	Stay the same	Increase

24. The enthalpy change for which of the following reactions would be equal to the enthalpy of formation for ethanol ($\text{CH}_3\text{CH}_2\text{OH}$)?

- a. $\text{CH}_3 + \text{CH}_2 + \text{OH} \rightarrow \text{CH}_3\text{CH}_2\text{OH}$
- b. $4 \text{C} + 6 \text{H}_2 + \text{O}_2 \rightarrow 2 \text{CH}_3\text{CH}_2\text{OH}$
- c. $2 \text{C} + 6 \text{H} + \text{O} \rightarrow \text{CH}_3\text{CH}_2\text{OH}$
- d. $2 \text{C} + 3 \text{H}_2 + \frac{1}{2} \text{O}_2 \rightarrow \text{CH}_3\text{CH}_2\text{OH}$

Unit 4: AP Free Response Practice #1 [2018 #1, shortened, 7 points]


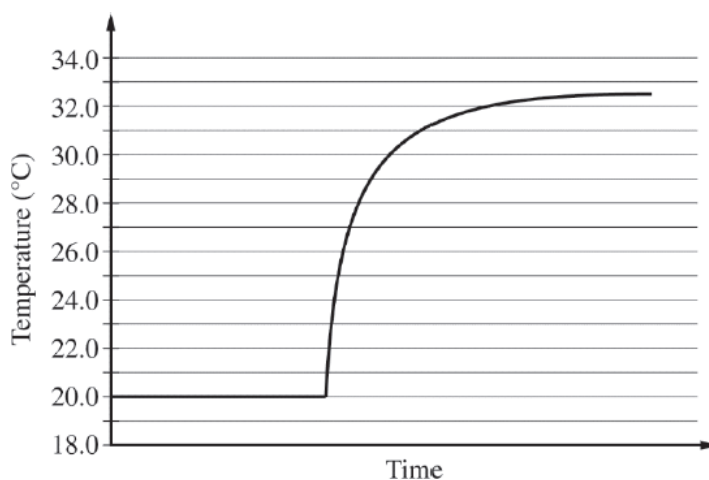
1. A student performs an experiment to determine the value of the enthalpy change, $\Delta H^\circ_{\text{rxn}}$, for the oxidation-reduction reaction represented by the balanced equation above.

In the experiment, the student uses the solutions shown in the table below.

Solution	Concentration (M)	Volume (mL)
$\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$	0.500	5.00
$\text{NaOCl}(\text{aq})$	0.500	5.00
$\text{NaOH}(\text{aq})$	0.500	5.00

- a. Using the balanced equation for the oxidation-reduction reaction and the information in the table above, determine which reactant is the limiting reactant. Justify your answer. [1 point]

The solutions, all originally at 20.0°C, are combined in an insulated calorimeter. The temperature of the reaction is monitored, as shown in the graph below.



- b. According to the graph, what is the temperature change of the reaction mixture? [1 point]

- c. The mass of the reaction mixture inside the calorimeter is 15.21 g.
- Calculate the magnitude of the heat energy, in joules, that is released during the reaction. Assume that the specific heat of the reaction mixture is $3.94 \text{ J/(g } ^\circ\text{C)}$ and that the heat absorbed by the calorimeter is negligible. [1 point]
 - Using the balanced equation for the oxidation-reduction reaction and your answer to part (a), calculate the value of the enthalpy change of the reaction, $\Delta H^\circ_{\text{rxn}}$, in $\text{kJ/mol}_{\text{rxn}}$. Include the appropriate algebraic sign with your answer. [2 points]

The student repeats the experiment, but this time doubling the volume of each of the reactants, as shown in the table below.

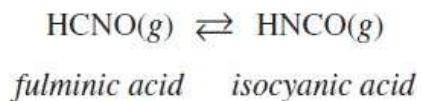
Solution	Concentration (M)	Volume (mL)
$\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$	0.500	10.0
$\text{NaOCl}(\text{aq})$	0.500	10.0
$\text{NaOH}(\text{aq})$	0.500	10.0

- The magnitude of the enthalpy change of the reaction, $\Delta H^\circ_{\text{rxn}}$, in $\text{kJ/mol}_{\text{rxn}}$, calculated from the results of the second experiment is the same as the result calculated in part (c)(ii). Explain this result. [1 point]
- Write the balanced net-ionic equation for the given reaction. [1 point]

Unit 4: AP Free Response Practice #2 [2017 #2, shortened, 5 points]

2. Answer the following questions about the isomers fulminic acid and isocyanic acid.

Fulminic acid can convert to isocyanic acid according to the equation below.



Fulminic Acid	Isocyanic Acid
$\text{H}-\text{C}\equiv\text{N}-\ddot{\text{O}}:$	$\text{H}-\ddot{\text{N}}=\text{C}=\ddot{\text{O}}:$

- a. Using the Lewis electron-dot diagrams of fulminic acid and isocyanic acid shown in the boxes above and the table of average bond enthalpies below, determine the value of ΔH° for the reaction of $\text{HCNO}(g)$ to form $\text{HNCO}(g)$. [2 points]

Bond	Enthalpy (kJ/mol)		Bond	Enthalpy (kJ/mol)		Bond	Enthalpy (kJ/mol)
N–O	201		C=N	615		H–C	413
C=O	745		C≡N	891		H–N	391

- b. A student claims that ΔS° for the reaction is close to zero. Explain why the student's claim is accurate. [1 point]
- c. Which species, fulminic acid (HCNO) or isocyanic acid (HNCO), is present in high concentration at equilibrium at 298 K? Justify your answer in terms of the thermodynamic favorability and the equilibrium constant. [2 points]

Unit 4: AP Free Response Practice #3 [LTF Free Response #2, 10 points]

3. The following questions relate to the synthesis reaction represented by the chemical equation above.
- Is the reaction thermodynamically favorable or unfavorable under standard conditions at 298 K? Justify your answer. [1 point]
 - In terms of the equilibrium constant, K , for the above reaction at 25°C
 - Predict whether K will be greater than, less than, or equal to one. Justify your choice. [1 point]
 - Calculate its value. [2 points]

- c. Given the following data, determine the ΔH° for the above reaction. [2 points]

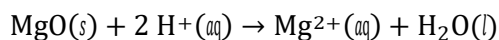
Substance	$\Delta H^\circ_{\text{ff}}$ (kJ mol ⁻¹)
NH ₃ (g)	-46.1

- d. In terms of the standard entropy change, ΔS°
- Predict the sign of ΔS° for the above reaction. Justify your answer. [1 point]

- ii. Calculate the value of ΔS° for the above reaction at 25°C. [1 point]

- e. Using the data in the table below and the enthalpy of reaction, $\Delta H^\circ_{\text{rxn}}$ determined in part (c), calculate the approximate bond energy of the nitrogen-hydrogen bond in ammonia. [2 points]

Bonds	Approximate Bond Energy (kJ mol ⁻¹)
N – H	???
H – H	430
N $\equiv$ N	960

Unit 4: AP Free Response Practice #4 [2013 FR #3, 10 points]

4. A student was assigned to the task of determining the enthalpy change for the reaction between solid MgO and aqueous HCl represented by the net ionic equation above. The student uses a polystyrene cup calorimeter and performs four trials. Data for each trial are shown in the table below.

Trial	Volume of 1.0 M HCl (mL)	Mass of MgO(s) Added (g)	Initial Temperature of Solution (°C)	Final Temperature of Solution (°C)
1	100.0	0.25	25.5	26.5
2	100.0	0.50	25.0	29.1
3	100.0	0.25	26.0	28.1
4	100.0	0.50	24.1	28.1

- a. Which is the limiting reactant in all four trials, HCl or MgO? Justify your answer. [1 point]
- b. The data in one of the trials is inconsistent with the data in the other three trials. Identify the trial with the inconsistent data and draw a line through the data from that trial in the table above. Explain how you identified the inconsistent data. [1 point]

For parts (c) and (d), use the data from one of the other three trials (i.e., not from the trial you identified in part (b) above.) Assume the calorimeter has a negligible heat capacity and that the specific heat of the contents of the calorimeter is 4.18 J/(g °C). Assume that the density of the HCl(aq) is 1.0 g/mL.

- c. Calculate the magnitude of q , the thermal energy change, when the MgO was added to the 1.0 M HCl(aq). Include units with your answer. [2 points]

- d. Determine the student's experimental value of ΔH° for the reaction between MgO and HCl in units of $\text{kJ/mol}_{\text{rxn}}$. [2 points]

- e. Enthalpies of formation for substances involved in the reaction are shown in table below. Using the information in the table, determine the accepted value of ΔH° for the reaction between MgO(s) and HCl(aq). [2 points]

Substance	ΔH_f° (kJ/mol)
MgO(s)	-602
H ₂ O(l)	-286
H ⁺ (aq)	0
Mg ²⁺ (aq)	-467

- f. The accepted value and the experimental value do not agree. If the calorimeter leaked heat energy to the environment, would it help account for the discrepancy between the values? Explain. [1 point]